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Project Name [Super Shop Management System/

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1. INTRODUCTION

Background to the Problem

Super shops are an important part of the retail business because they provide many types of products in one place. As the number of customers and products increases, managing a super shop becomes more difficult. Many small and medium-sized super shops still use manual records or basic software to manage stock, billing, sales, employee information, and customer records. These methods can cause mistakes, take more time, and make it difficult to keep accurate records. Manual management also creates other problems. Employees need more time to update stock information, prepare bills, and keep sales records. Sometimes products may go out of stock without anyone noticing, while other products may be overstocked. Billing mistakes and incorrect stock records can also affect customer satisfaction and reduce the overall efficiency of the business. Preparing sales reports manually is time-consuming and does not always provide accurate information for making business decisions.

The proposed Super Shop Management System is designed to make these daily tasks easier. It helps manage products, inventory, billing, sales, customer information, employee information, and reports in one system. The system automatically updates stock after every sale, stores sales records, and generates bills quickly and accurately. It also provides reports that help the store management understand sales performance and inventory status. The system is also designed to improve communication between different users of the super shop. Each user has a specific role and can access only the features related to their work. This helps reduce confusion, improves security, and allows employees to complete their tasks more efficiently.

As a result, the daily operations of the super shop become more organized, and both employees and customers benefit from a faster and more reliable service. By reducing manual work and keeping records up to date, the system saves time, reduces errors, and helps the shop run more efficiently. It also provides accurate information that supports better business decisions. Overall, the system improves daily operations, increases productivity, and provides better service to customers.

The system is intended for the following target users:

Admin:

- **Manages User Accounts:** Creates, updates, and removes user accounts and assigns user roles.
- **System Management:** Maintains system settings and ensures smooth system operation.
- **Maintains Security:** Controls user access, secures system data, and performs regular backups.
- **Generate Reports:** Monitors system activities and generates management reports.

Store Manager:

- **Monitors Daily Operations:** Oversees the daily activities of the super shop.
- **Manages Employees:** Assigns tasks and monitors employee performance.

- **Monitors Sales:** Reviews sales performance and identifies areas for improvement.
- **Reviews Reports:** Checks sales and inventory reports to support business decisions.

Inventory Manager:

- **Manages Products:** Adds, updates, and removes product information.
- **Maintains Inventory:** Monitors stock levels and keeps inventory records up to date.
- **Updates Stock:** Updates stock quantities after sales and new deliveries.
- **Handles Restocking:** Monitors low-stock items and arranges stock replenishment.

Cashier:

- **Processes Sales:** Scans products and completes customer purchases.
- **Generates Bills:** Calculates totals and prepares customer bills.
- **Processes Payments:** Accepts cash, card, and digital payments.
- **Issues Receipts:** Prints receipts and handles product returns when required.

Customer

- **Purchases Products:** Selects and buys products.
- **Browse Product:** Views product information, prices, and availability.
- **Makes Payments:** Completes payments using the available payment methods.

2. SCOPES & FEATURES

2.1 User Registration and Secure Login

The system allows new customers to create an account by providing the required personal information. After successful registration, users can access the system using their username and password. This feature ensures that only registered users can perform customer-related activities.

2.2 User Login

The system allows registered users to log in using their username and password. It verifies the user information before granting access to the system. This helps protect system data and ensures that only authorized users can access the available features.

2.3 Role-Based Access Control

The system provides different access permissions for different types of users. Each user can access only the features related to their responsibilities. This improves system security and helps users perform their tasks more efficiently.

2.4 User Account Management

The system allows authorized users to create, update, and remove user accounts. It also provides the option to assign user roles according to job responsibilities. This helps keep user information organized and ensures proper system access.

2.5 Password Management

The system allows users to change their passwords whenever required. The current password must be verified before a new password is accepted. This feature improves account security and protects user information.

2.6 Product Management

The system allows products to be added, updated, and removed whenever necessary. It stores important product information such as name, price, category, and stock quantity. This helps keep the product catalog accurate and up to date.

2.7 Product Search

The system allows users to search for products using their names or other available information. It displays matching products quickly and helps users find the required items easily. This feature saves time and improves the overall shopping experience.

2.8 Product Browsing

The system allows customers to browse the available products in the store. Product information, prices, and availability are displayed clearly. This helps customers choose products more easily before making a purchase.

2.9 Product Details

The system provides detailed information about each product. Customers can view the product description, price, and stock availability before purchasing. This helps customers make better purchasing decisions.

2.10 Inventory Management

The system maintains inventory records for all available products. It keeps track of stock quantities and updates inventory information regularly. This helps reduce stock management errors and improves inventory control.

2.11 Stock Update

The system automatically updates stock quantities after every sale and product return. This keeps inventory records accurate without requiring manual updates. It also helps management monitor product availability more effectively.

2.12 Low Stock Monitoring

The system monitors inventory levels and identifies products that are running low. It helps management arrange product restocking before items become unavailable. This improves inventory planning and customer service.

2.13 Shopping Cart Management

The system allows customers to add products to their shopping cart before placing an order. Customers can also remove products or change their selected items. This provides flexibility during the purchasing process.

2.14 Order Placement

The system allows customers to place orders for the selected products. After the order is confirmed, the purchase information is stored in the system. This helps complete the purchasing process quickly and accurately.

2.15 Billing System

The system generates bills automatically after products are selected for purchase. It calculates the total amount and prepares the bill quickly and accurately. This reduces billing errors and provides faster customer service.

2.16 Payment Processing

The system supports different payment methods such as cash, card, and digital payment. It records payment information after a successful transaction. This helps complete purchases securely and efficiently.

2.17 Receipt Generation

The system generates a receipt after every successful payment. The receipt contains the details of the purchased products and the total payment amount. This provides customers with proof of purchase and maintains transaction records.

2.18 Product Return Management

The system records product return information whenever a customer returns an item. It automatically updates the inventory after the return is completed. This helps maintain accurate stock records and improves customer service.

2.19 Sales Management

The system records all sales transactions performed in the store. It stores sales information for future references and reporting. This helps management monitor business performance and daily sales activities.

2.20 Report Generation

The system generates different types of reports, including sales reports, inventory reports, and management reports. These reports provide accurate information about business activities. They help management monitor performance and make better business decisions.

2.21 Employee Task Management

The system allows employee tasks to be assigned and monitored. It helps organize daily work and improves communication among employees. This makes daily store operations more efficient.

2.22 Inventory Request Management

The system allows inventory requests to be submitted and approved when products need to be restocked. It helps maintain sufficient stock levels for customer demand. This reduces the possibility of products becoming unavailable.

2.23 Purchase History

The system stores customer purchase records for future references. Customers can view their previous orders whenever required. This helps customers keep track of their purchases and improves the overall shopping experience.

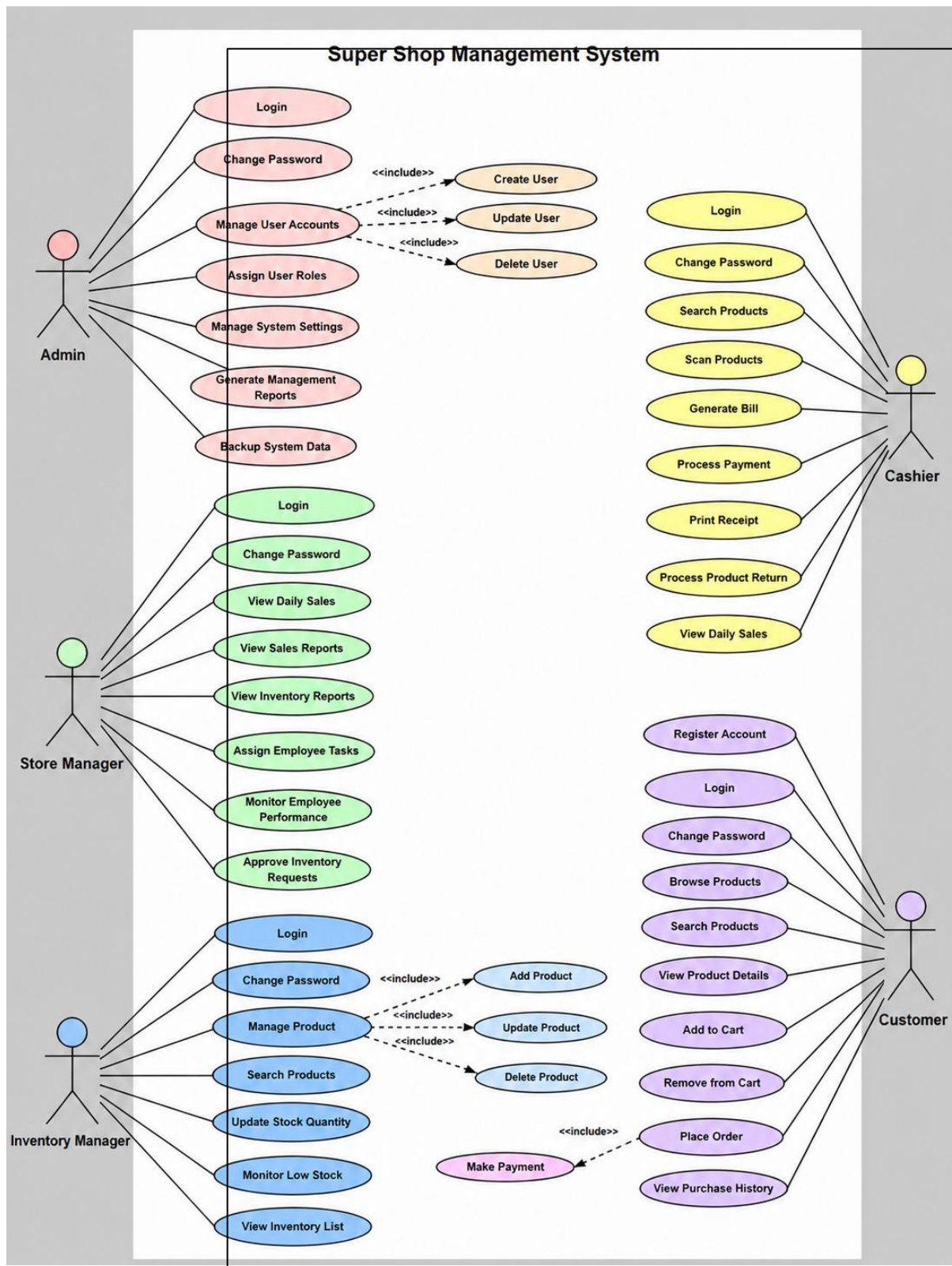
2.24 System Backup

The system performs regular backups of important business information. Backup files help recover data in case of accidental loss or system failure. This improves data reliability and supports continuous system operation.

2.25 System Security and Monitoring

The system monitors important activities such as user login, product updates, sales transactions, and account modifications. It helps maintain system security and keeps business records organized. This also supports better system management and improves overall reliability.

3. USE CASE DIAGRAM



4. USER STORY TABLE

As A/An	I want to	So that	Acceptance Criteria
Admin	Login with username and password	I can securely access the system	Valid credentials allow access to the Admin Dashboard
	Create a new user account	New employees can use the system	User information is saved and a success notification is shown
	Update user information	User information remains accurate	Updated information is displayed immediately
	Delete a user account	Unauthorized users cannot access the system	Confirmation message is shown before deletion
	Assign user roles	Every user has proper system access	Selected role is assigned successfully
	Manage system settings	The system works according to business requirements	Updated settings are saved successfully
	Generate management reports	I can monitor overall business performance	Reports are generated with accurate data
	Backup system data	Important information is protected	Backup completes successfully with confirmation
	Change password	My account becomes more secure	Old password and new password validation is required
Store Manager	Login with username and password	I can access my dashboard securely	Successful authentication displays the dashboard
	View daily sales	I can monitor store performance	Sales summary is displayed
	View inventory reports	I know the current stock status	Inventory report is generated
	Assign tasks to employees	Daily operations run efficiently	Assigned tasks appear in employee accounts
	Monitor employee performance	I can evaluate employees	Performance report is displayed
	View sales reports	I can make business decisions	Monthly and yearly sales reports are available
	Approve inventory requests	Stock shortages are handled quickly	Approval updates the inventory request status
	Change password	My account remains secure	Password is updated after validation
Inventory Manager	Login with username and password	I can manage inventory securely	Dashboard appears after successful login

	Add a new product	New products become available for sale	Product is saved successfully
	Update product information	Customers see accurate product details	Updated product information is displayed
	Remove a product	Unavailable products are removed	Product disappears after confirmation
	Search products	I can quickly find product information	Matching products are displayed
	Update stock quantity	Inventory stays accurate	Stock quantity changes successfully
	Monitor low-stock items	Products can be restocked on time	Low-stock notification appears
	View inventory list	I can monitor all available products	Product list displays current stock
	Change password	My account stays protected	Password is updated successfully
Cashier	Login with username and password	I can process customer purchases securely	Dashboard opens after successful login
	Search products	I can quickly find customer items	Product search returns matching results
	Scan products	Billing becomes faster and more accurate	Selected products are added to the bill
	Generate bills	Customers receive correct invoices	Bill displays products, quantity, and total
	Process payments	Customers can complete purchases	Cash, card, and digital payments are accepted
	Print receipts	Customers receive proof of purchase	Receipt is generated successfully
	Process product returns	Returned products are recorded correctly	Return transaction updates inventory
	View today's sales	I can track my daily performance	Daily sales report is displayed
Customer	Change password	My account remains secure	Password changes after successful verification
	Register a new account	I can purchase products easily	Registration completes successfully
	Login with username and password	I can access my account	Dashboard opens after authentication

	Browse products	I can view available products	Product catalog is displayed
	Search products	I can quickly find desired products	Matching products are listed
	View product details	I can make informed purchasing decisions	Product information, price, and availability are shown
	Add products to cart	I can purchase multiple items together	Cart updates with selected products
	Remove products from cart	I can modify my order	Cart updates correctly
	Place an order	I can buy selected products	Order is created successfully
	Make payment	My purchase is completed	Payment confirmation is displayed
	View purchase history	I can review previous orders	Order history is displayed
	Change password	My account becomes more secure	Password is updated after verification

5. REQUIREMENT TRACEABILITY MATRIX

5.1 Functional Requirements

- User registration and login.
- Role-based access for Admin, Store Manager, Inventory Manager, Cashier, and Customer.
- Admin can create, update, delete user accounts, assign user roles, and manage system settings.
- Store Manager can monitor daily sales, review inventory reports, assign employee tasks, and approve inventory requests.
- Inventory Manager can add, update, remove, and search products, update stock quantities, and monitor low-stock items.
- Cashiers can scan products, generate bills, process payments, print receipts, and process product returns.
- Customers can browse products, search products, view product details, manage shopping carts, place orders, and make payments.
- The system automatically updates inventory after every sale or product return.
- Users can view reports, inventory status, sales history, and purchase history according to their roles.

5.2 Non-Functional Requirements

- The system should respond to user requests within 2–3 seconds under normal operating conditions.
- User passwords must be securely encrypted.
- Only authorized users should be able to access features based on their assigned roles.
- The system should automatically update inventory records after sales and returns.
- The user interface should be simple, intuitive, and easy to use.
- The system should support future expansion, including additional users, products, and store branches, without major changes.
- The code should be modular, maintainable, and well-documented for future development.
- The system should maintain activity logs for important operations such as login attempts, product updates, sales transactions, payment processing, and user account modifications.
- The system should perform regular backups to protect business data from accidental loss.

